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Кафедра интеллектуальных информационных технологий

Лабораторная работа №3

По дисциплине «Обработка изображений в интеллектуальных системах»

Тема: «Конструирование моделей на базе предобученных нейронных сетей»

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Цель работы: осуществлять обучение нейросетевого детектора для решения задачи обнаружения заданных объектов.

Общее задание

1. Базируясь на своем варианте, ознакомится с выборкой для обучения детектора, выполнить необходимые преобразования данных для организации процесса обучения (если это нужно!);
2. Для заданной архитектуры нейросетевого детектора организовать процесс обучения для своей выборки. Оценить эффективность обучения на тестовой выборке (mAP);
3. Реализовать визуализацию работы детектора из пункта 1 (обнаружение знаков на отдельных фотографиях из сети Интернет);
4. Оформить отчет по выполненной работе, залить исходный код и отчет в соответствующий репозиторий на github.

Задание по вариантам

Вариант	Выборка	Оптимизатор
17	YOLO26m	**Счетчики расхода воды:** https://universe.roboflow.com/koer3741-gmail-com/watermeteramrv2/dataset/1

Ход работы

Код программы:

```
import glob
import random
import zipfile
from pathlib import Path

import matplotlib.patches as patches
import matplotlib.pyplot as plt
import pandas as pd
import torch
import yaml
from PIL import Image
from ultralytics import YOLO

random.seed(42)
print("GPU доступен:", torch.cuda.is_available())

ZIP_PATH = "/content/WaterMeterAMRV2-1.zip"
DATASET_DIR = Path("/content/dataset")

with zipfile.ZipFile(ZIP_PATH) as z:
    z.extractall(DATASET_DIR)
print("Содержимое датасета:", [p.name for p in DATASET_DIR.iterdir()])

# 2. Проверка и настройка data.yaml

yaml_path = DATASET_DIR / "data.yaml"
with open(yaml_path) as f:
    data_cfg = yaml.safe_load(f)
print("Классы:", data_cfg["names"], "| число классов:", len(data_cfg["names"]))

data_cfg["path"] = str(DATASET_DIR)
data_cfg["train"] = "train/images"
data_cfg["val"] = "valid/images"
data_cfg["test"] = "test/images"
with open(yaml_path, "w") as f:
    yaml.safe_dump(data_cfg, f, allow_unicode=True)

for split in ["train", "valid", "test"]:
    folder = DATASET_DIR / split / "images"
    n = len(list(folder.glob("*"))) if folder.exists() else 0
    print(f"{split:>6}: {n} изображений")
```

```
def show_samples(split="train", n=6):
    img_paths = random.sample(list((DATASET_DIR / split / "images").glob("*")), n)
    fig, axes = plt.subplots(2, 3, figsize=(14, 8))
    for ax, p in zip(axes.flatten(), img_paths):
        img = Image.open(p).convert("RGB")
        W, H = img.size
        ax.imshow(img)
        ax.axis("off")
        lbl = DATASET_DIR / split / "labels" / (p.stem + ".txt")
        if lbl.exists():
            for line in lbl.read_text().strip().splitlines():
                parts = line.split()
                if len(parts) < 5:
                    continue
                _, cx, cy, w, h = map(float, parts[:5])
                x0, y0 = (cx - w / 2) * W, (cy - h / 2) * H
                ax.add_patch(patches.Rectangle((x0, y0), w * W, h * H,
                                               fill=False, edgecolor="lime", linewidth=2))
    plt.suptitle("Примеры из обучающей выборки (зелёным показана разметка)")
    plt.tight_layout()
    plt.show()


show_samples("train")
model = YOLO("yolo26m.pt")
EPOCHS = 30
results = model.train(
    data=str(yaml_path),
    epochs=EPOCHS,
    imgsz=640,
    batch=16,
    project="runs_meters",
    name="yolo26m",
    exist_ok=True,
    seed=42,
)
save_dir = Path(model.trainer.save_dir)
print("Результаты обучения в:", save_dir)

df = pd.read_csv(save_dir / "results.csv")
df.columns = df.columns.str.strip()

fig, axes = plt.subplots(1, 2, figsize=(13, 5))
axes[0].plot(df["epoch"], df["train/box_loss"], label="Train box loss", marker="o", markersize=3)
axes[0].plot(df["epoch"], df["val/box_loss"], label="Val box loss", marker="o", markersize=3)
axes[0].set_xlabel("Эпоха")
axes[0].set_ylabel("Loss")
axes[0].set_title("YOLO26m: изменение ошибки")
axes[0].legend()
axes[0].grid(alpha=0.3)

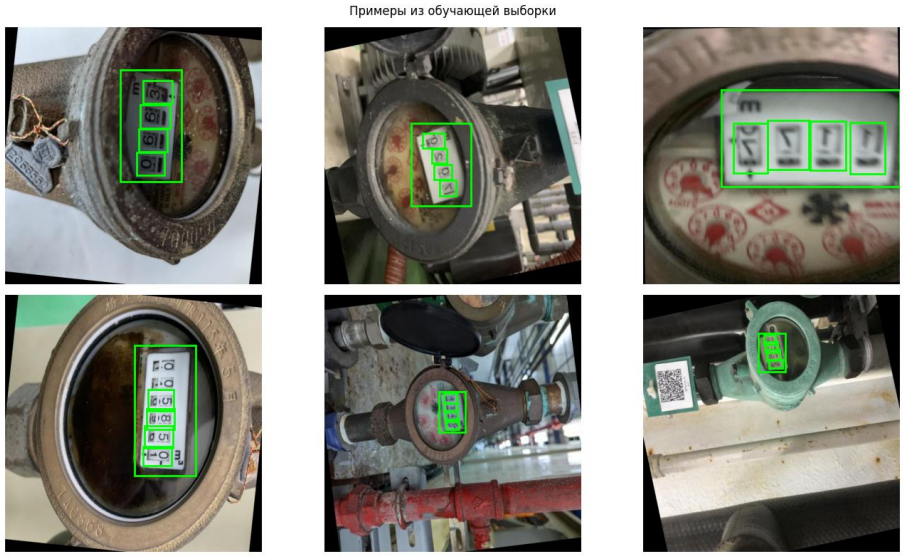
axes[1].plot(df["epoch"], df["metrics/mAP50(B)"], label="mAP@0.5", marker="o", markersize=3)
axes[1].plot(df["epoch"], df["metrics/mAP50-95(B)"], label="mAP@0.5:0.95", marker="o", markersize=3)
axes[1].set_xlabel("Эпоха")
axes[1].set_ylabel("mAP")
axes[1].set_title("YOLO26m: изменение mAP на валидации")
axes[1].legend()
axes[1].grid(alpha=0.3)
plt.tight_layout()
plt.show()

best = YOLO(str(save_dir / "weights" / "best.pt"))
metrics = best.val(data=str(yaml_path), split="test", imgsz=640)
print("Результаты на тестовой выборке:")
print(f"Precision: {metrics.box.mp:.4f}")
print(f"Recall: {metrics.box.mr:.4f}")
print(f"mAP@0.5: {metrics.box.map50:.4f}")
print(f"mAP@0.5:0.95: {metrics.box.map:.4f}")
```

```
def detect_and_show(sources, conf=0.25, imgsz=640):
    for src in sources:
        img = Image.open(src).convert("RGB")
        res = best.predict(img, conf=conf, imgsz=imgsz, verbose=False)[0]
        plt.figure(figsize=(9, 6))
        plt.imshow(res.plot()[..., ::-1])
        plt.axis("off")
        n = len(res.boxes)
        confs = ", ".join(f"{c:.2f}" for c in res.boxes.conf.tolist())
        plt.title(f"Найдено объектов: {n}" + (f" (уверенность: {confs})" if n else ""))
        plt.show()


photos = glob.glob("/content/test/*")
print(photos)
detect_and_show(photos, conf=0.25, imgsz=1280)
```

Вывод программы:




```
           Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            1/30       8.8G      1.565      3.284    0.00526         40        640: 100% ▕▕▕
```

```
            5/30      8.99G      1.488     0.7549   0.005016         51        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:54
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.2it/s 6.0s
                         all        389       2059      0.924      0.896      0.945      0.542

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            6/30         9G      1.464     0.7091   0.004937         70        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:54
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.2it/s 6.0s
                         all        389       2059      0.938      0.919      0.961      0.544

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            7/30         9G      1.445     0.6531    0.00483         57        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:54
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.2it/s 5.9s
                         all        389       2059      0.913      0.913      0.941      0.482

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            8/30         9G      1.426     0.6422   0.004756         51        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:54
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.1it/s 6.3s
                         all        389       2059      0.925      0.922      0.959      0.542

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            9/30       9.03G      1.403     0.6119   0.004608         59        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:53
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.2it/s 5.8s
                         all        389       2059      0.949      0.944      0.971      0.571

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            10/30      9.01G      1.397     0.5885   0.004615         42        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:53
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.1it/s 6.3s
                         all        389       2059      0.942       0.94      0.964       0.57

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            11/30      9.01G       1.38     0.5916   0.004486         32        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:54
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.2it/s 5.8s
                         all        389       2059      0.941      0.928      0.961      0.531

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            12/30      9.01G      1.356     0.5594    0.00447         49        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:53
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.2it/s 6.0s
                         all        389       2059      0.954      0.937      0.974      0.577

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            13/30        9G      1.319     0.5402   0.004254         27        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:53
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.1it/s 6.2s
                         all        389       2059      0.955      0.939      0.974       0.58

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            14/30      9.02G      1.326     0.5365   0.004272         29        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:53
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.0it/s 6.4s
                         all        389       2059      0.948      0.941      0.975      0.592
```

```
            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            15/30      9.02G      1.291     0.5147   0.004236         78        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:53
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.0it/s 6.4s
                         all        389       2059       0.96      0.953       0.98      0.595

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            16/30      9.01G      1.287     0.5015   0.004085         74        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:54
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.0it/s 6.4s
                         all        389       2059       0.96       0.95      0.979      0.582

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            17/30      9.05G      1.265     0.4913   0.004035         45        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:53
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.2it/s 5.9s
                         all        389       2059      0.958      0.961      0.978      0.602

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            18/30      9.01G      1.242     0.4658   0.003958         60        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:55
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.1it/s 6.1s
                         all        389       2059      0.958      0.952      0.982      0.605

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            19/30        9G      1.234     0.4548   0.003923         61        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:55
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.0it/s 6.5s
                         all        389       2059      0.965       0.95      0.978      0.616

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            20/30      9.02G      1.217     0.4545   0.003844         31        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:56
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.1it/s 6.2s
                         all        389       2059      0.964       0.96      0.984      0.617
      Closing dataloader mosaic
      albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01,
method='weighted_average', num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8,
8))

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            21/30      8.99G       1.19     0.3608   0.004508         30        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:56
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.0it/s 6.6s
                         all        389       2059      0.967      0.951      0.979      0.615

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            22/30      8.99G      1.161       0.35   0.004382         31        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:54
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.0it/s 6.4s
                         all        389       2059       0.96      0.954      0.978      0.609

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
            23/30        9G      1.144     0.3431   0.004284         28        640: 100% ▒▒▒▒▒▒▒▒▒▒▒▒
218/218 1.9it/s 1:54
                       Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
▒▒▒▒▒▒▒▒▒▒▒▒▒▒ 13/13 2.1it/s 6.2s
                         all        389       2059      0.971      0.949      0.981      0.617

            Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
```

```
           24/30         9G      1.125     0.3283   0.004143         32        640: 100% ━━━━━━━━━━━━
218/218 1.9it/s 1:54
                     Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 13/13 1.9it/s 6.9s
                       all        389       2059      0.971      0.953      0.983      0.628

           Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
           25/30         9G      1.102     0.3229   0.004014         32        640: 100% ━━━━━━━━━━━━
218/218 1.9it/s 1:55
                     Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 13/13 2.1it/s 6.1s
                       all        389       2059      0.972      0.957      0.982       0.63

           Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
           26/30         9G      1.094     0.3184   0.003959         30        640: 100% ━━━━━━━━━━━━
218/218 1.9it/s 1:54
                     Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 13/13 1.9it/s 6.7s
                       all        389       2059      0.972       0.96      0.983      0.631

           Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
           27/30      8.99G      1.079     0.3091   0.003931         32        640: 100% ━━━━━━━━━━━━
218/218 1.9it/s 1:54
                     Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 13/13 2.1it/s 6.1s
                       all        389       2059      0.962      0.966      0.983      0.628

           Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
           28/30      8.99G      1.073     0.3084   0.003882         33        640: 100% ━━━━━━━━━━━━
218/218 1.9it/s 1:54
                     Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 13/13 1.9it/s 6.8s
                       all        389       2059      0.971      0.958      0.984      0.629

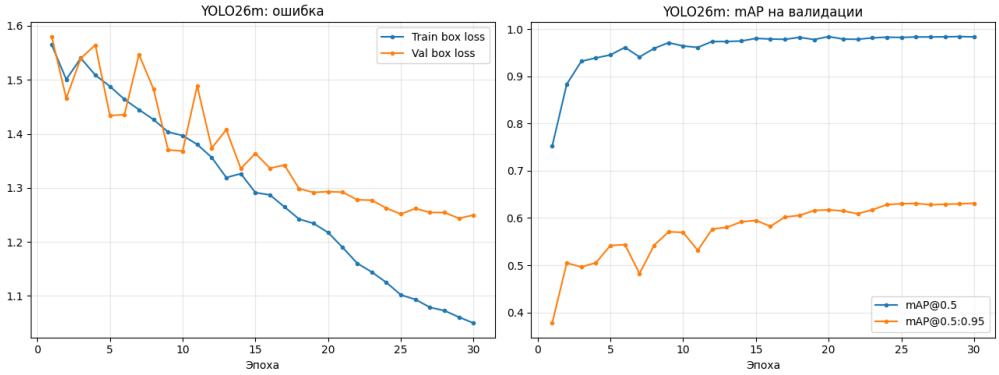
           Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
           29/30      8.99G      1.061     0.3026   0.003824         30        640: 100% ━━━━━━━━━━━━
218/218 1.9it/s 1:54
                     Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 13/13 2.1it/s 6.3s
                       all        389       2059      0.972       0.96      0.984       0.63

           Epoch    GPU_mem   box_loss   cls_loss    l1_loss  Instances       Size
           30/30         9G       1.05     0.3007   0.003797         33        640: 100% ━━━━━━━━━━━━
218/218 1.9it/s 1:54
                     Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 13/13 2.0it/s 6.3s
                       all        389       2059      0.968      0.964      0.983      0.631

     30 epochs completed in 1.042 hours.
     Optimizer stripped from /content/runs/detect/runs_meters/yolo26m/weights/last.pt, 44.1MB
     Optimizer stripped from /content/runs/detect/runs_meters/yolo26m/weights/best.pt, 44.1MB

     Validating /content/runs/detect/runs_meters/yolo26m/weights/best.pt...
     Ultralytics 8.4.175 🚀 Python-3.13.15 torch-2.11.0+cu130 CUDA:0 (Tesla T4, 14913MiB)
     YOLO26m summary (fused): 130 layers, 20,358,704 parameters, 0 gradients, 68.1 GFLOPs
                     Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 13/13 1.4it/s 9.3s
                       all        389       2059      0.968      0.965      0.983      0.632
                         0        151        196      0.898      0.896      0.936       0.57
                         1        122        161       0.92      0.933      0.966      0.607
                         2         97        106       0.99      0.961      0.994      0.589
                         3        124        148      0.986      0.973      0.991      0.656
                         4        148        184      0.988      0.973      0.981      0.629
                         5         96        111      0.939      0.963      0.989      0.592
                         6        125        151          1      0.987      0.991      0.651
                         7         98        124      0.992      0.962      0.985      0.626
                         8         60         67       0.97      0.961      0.993      0.601
                         9         58         60      0.965      0.983      0.989      0.614
                   counter        367        373      0.993      0.995      0.995      0.851
```

```
                liters        375        378      0.979      0.992       0.99      0.596
Speed: 1.3ms preprocess, 13.8ms inference, 0.0ms loss, 2.1ms postprocess per image
Results saved to /content/runs/detect/runs_meters/yolo26m
/content/runs/detect/runs_meters/yolo26m
```




```
     Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ▒▒▒▒▒▒▒▒▒▒▒▒ 25/25
1.7it/s 14.3s
                   all        389       2105      0.967      0.948      0.972      0.609
                     0        162        213       0.93      0.915      0.948      0.568
                     1        108        153      0.923      0.944      0.963       0.54
                     2         87         90      0.967      0.971      0.969      0.552
                     3        125        154      0.979       0.92      0.955      0.591
                     4        132        165      0.988      0.984      0.993      0.636
                     5        119        167      0.953      0.962      0.988      0.593
                     6        112        130      0.949      0.946      0.977      0.615
                     7        100        115      0.974       0.96      0.993      0.619
                     8         64         75          1      0.878      0.949      0.571
                     9         70         75       0.98       0.92      0.945      0.583
               counter        371        383      0.988       0.99      0.993      0.834
                liters        377        385      0.975      0.984      0.989      0.602
Speed: 2.3ms preprocess, 26.9ms inference, 0.0ms loss, 1.9ms postprocess per image
Results saved to /content/runs/detect/val
Precision: 0.9671
Recall: 0.9479
mAP@0.5: 0.9717
mAP@0.5:0.95: 0.6090
```

Пример работы на моем счетчике:



Вывод: осуществила обучение нейросетевого детектора для решения задачи обнаружения заданных объектов.